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MICRO-CAMERA BASED AIROTOR

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Abstract: With the advancement of technology the present invention related to the dental airtor including micro-camera for capturing and recording. This airtor also going to have LED light which can incorporated into hand-piece for improvement of visibility and comfort of working. Wireless transmission of capture image or live feeding of treatment site to destination device so that it will increase the operability of the dentist during treatment. In general this airtor are used for various purpose such as Oral cleaning. Tooth decay and plug removal. Due to high speed the friction creates significant heat within bur and with help of cooling system can be prevent. With this innovative technique those hard to reach out and hard to see area are going to be more accessible

Introduction : A dental airtor is a hand-held, mechanical instrument used to perform a variety of common dental procedures, including removing decay, polishing fillings, performing cosmetic dentistry, and altering prostheses. The airtor itself consists of internal mechanical components which initiate a rotational force and provide power to the cutting instrument, usually a dental burr. The type of apparatus used clinically will vary depending on the required function dictated by the dental procedure. It is common for a light source and cooling water-spray system to also be incorporated into certain airtor. But in a dental airtor, treatment may take longer amount of time to perform the task due to limitations of viewing angles. So by using micro-camera in a dental airtor we can reduce the limitations of viewing angles and it will also help to perform treatment with much accuracy, visibility and overall success of the procedure



(a) Airtor



(b) Display



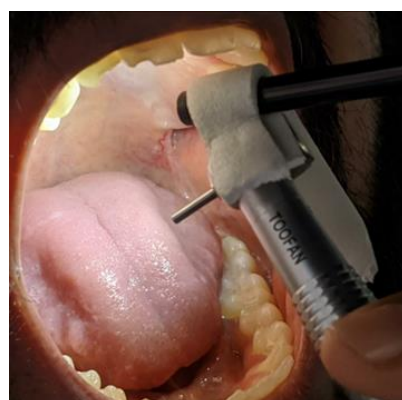
(c) Endoscopic Camera

Results:

Application Output Window



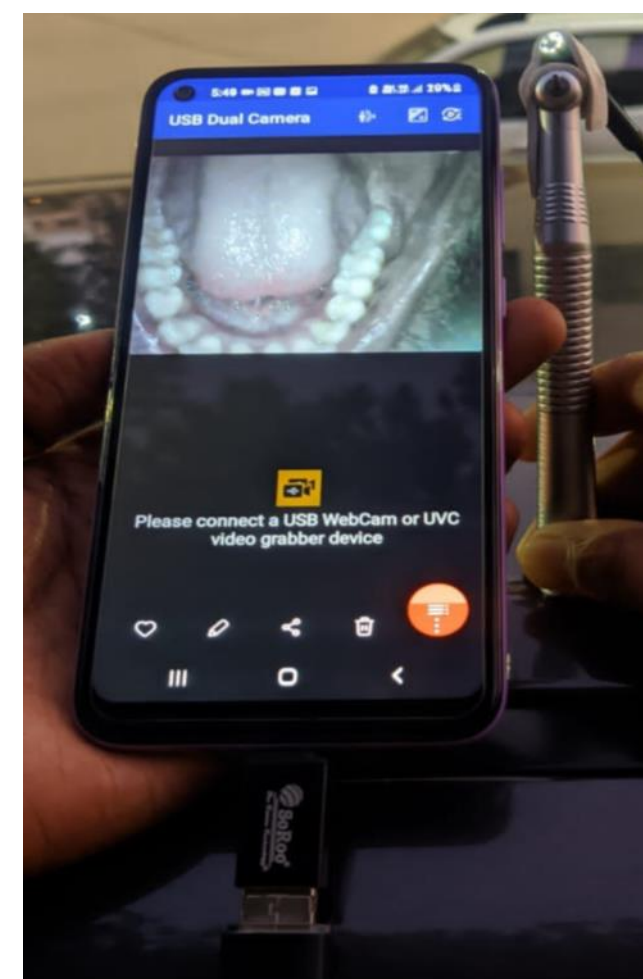
Demonstration



Project Model



Display Output



Conclusion and Future scope:

Conclusion: In using the present invention, the dentist will place the monitor in a convenient viewing location and turn on the master switch and the LED. Using the micro camera based airtor, on which the camera viewing attachment is mounted, the dentist will activate the excavating bur in order to remove decay from a patient's cavity. As the decay is removed from the cavity, the dentist is able to see a magnified, real-time image .

Future scope:

- With the help of camera based airtor we will be to fit camera in handpiece so it will enable the possibilities of using machine learning capabilities to detect infections in cavities
- An anti-fogging mechanism which blow air toward a front side of the light receiving port can also be added to take the high resolution photos while performing operation itself while using water gun of airtor.

References: 1]Hand-piece with built-in camera and dental treatment method using said hand-piece. 2012-08-31 <https://patents.google.com/patent/EP2891467A1/en>
[2] Dental bur cooling system 1989-10-16
<https://patents.google.com/patent/US5096421A/en>

Program Outcome:

PO5,PO9,PO10,PO11,PO12,PS01